

During the previous sprint, my key individual contribution was integrating our registration system with Firebase, transitioning the app from local data storage to a centralized, secure authentication system. Specifically, I was responsible for linking the code to Firebase by configuring the necessary dependencies, including `firebase_auth: 6.1.0`, `firebase_storage`, and `cloud_firestore: 4.1.1`, and ensuring they were correctly set up in `pubspec.yaml`. I also worked with the `firebase_options.dart` file, which contains the unique project settings needed for Firebase initialization. On the coding side, I revised `Registration.dart` so that user data — first name, last name, desired username, email, and date of birth — could be reliably recorded in Firebase. I also implemented logic to check for duplicate registrations, providing feedback in the UI when a user attempts to register with the same information.

One of the main challenges I encountered was setting up the development environment. I had previously used my MacBook terminal for coding projects and PuTTY to write code on the school computer, so switching to VS Code and configuring Flutter with all the required files in a single, working environment was difficult. I wanted to start coding immediately, but without the correct configuration, I couldn't run or test the project effectively. Another significant challenge was managing dependencies. Some versions I initially installed were incompatible with each other or with Flutter itself, which caused errors and prevented the app from running. To resolve this, I had to research online resources extensively to find the latest compatible versions and carefully update the project. This experience taught me the importance of checking version compatibility and understanding how dependencies interact in a Flutter project.

Through this process, I learned several new skills. I became proficient in configuring Firebase for Flutter, understanding how `firebase_options.dart` connects the app to the correct Firebase project, and implementing authentication logic. I also improved my debugging and problem-solving skills, particularly in managing dependencies and development environment setup.

Looking forward, there are areas for improvement. Personally, I could anticipate potential setup and dependency issues earlier and document the correct configuration to save time. From a team perspective, more structured testing and collaborative reviews could help catch configuration or logic errors sooner. In the next sprint, I plan to focus on connecting the registration page with the login page and implementing a unique username policy. Overall, this sprint was a valuable learning experience that strengthened my technical skills and allowed me to make a meaningful contribution to the team's progress.